

Lessons Learned

Introduction

This project provided hands-on experience in building a simulated cybersecurity environment and understanding how real-world attack detection works.

Through this process, several important technical and practical lessons were learned.

Technical Learnings

Importance of Logging

Sysmon provides detailed visibility into system activities

Process creation and command-line logs are critical for detection

Without proper logging, attacks cannot be detected

SIEM (Splunk) Usage

Learned how to search and analyze logs using SPL queries

Understood how to create alerts and dashboards

Realized the importance of centralized log monitoring

Detection Engineering

Built detection rules based on system behavior

Understood how attackers use PowerShell and CMD

Learned to detect suspicious patterns instead of relying on signatures

Network Monitoring

Learned how network connections are logged (Event ID 3)

Understood how attackers may communicate with external systems

Practical Learnings

Think Like an Attacker

Understanding attacker behavior helps improve detection

Simulating attacks gives better insight into threats

Detection is More Important Than Prevention

It is not always possible to stop attacks

Early detection is critical for response

Real-Time Monitoring

Continuous monitoring is essential in cybersecurity

Dashboards help visualize system activity effectively

Challenges Faced

Difficulty in configuring Splunk and forwarder initially

Troubleshooting log forwarding issues

Understanding Sysmon event logs

Writing effective detection queries

Improvements for Future

Add advanced attack simulations (lateral movement, persistence)

Improve detection rules with better filtering

Integrate threat intelligence feeds

Automate detection and response

Key Takeaway

Cybersecurity is not just about tools — it is about:

- Visibility
- Analysis
- Detection
- Continuous improvement

Final Reflection

This project helped build a strong foundation in:

- SOC operations
- Log analysis
- Threat detection

It provided practical experience that aligns with real-world cybersecurity roles such as SOC Analyst and Security Engineer.